

Updating the model for the medium-term projection of administered prices

This box updates the medium-term projection model for administered prices. The equations that represent the dynamics of the 24 items that make up the administered prices basket are presented, along with more detailed explanations for the items with the greatest weight. The model's properties are illustrated through impulse response functions to shocks in the exchange rate, oil prices, market prices, and the Assistance Expenses Value Index (IVDA).

This box presents the current specification of the model for generating projections and medium-term scenarios for administered prices.¹ This model is part of the set of tools used by Copom to project inflation. In particular, the inflation projections and scenarios from the semi-structural models are formed considering the set of equations from the administered prices model. Hence, the projections generated for headline inflation incorporate the feedback effects between market and administered prices.

Administered prices refer to the components of the Extended National Consumer Price Index (IPCA) whose adjustments are determined by contracts, public bodies, or regulatory agencies, such as electricity tariffs and health insurance plans. Fuel prices are also considered administered prices.² In general, administered prices are less sensitive to economic market conditions and introduce greater inertia into the behavior of inflation due, in many cases, to the presence in their contracts and regulations of clauses indexing them to past inflation. In addition, they are an important pass-through channel for exchange rate and oil price changes.

The BCB's inflation projections are based on a combination of medium-term model projections and short-term expert projections.³ Experts have specific information that is relevant to short-term dynamics but is difficult to incorporate properly into macroeconomic models. In general, experts' projections for the short term are built using a bottom-up approach, i.e. projecting variables at a higher level of disaggregation and then aggregating them. These projections also incorporate knowledge of the methodological details of data compilation. In the case of administered prices, knowledge of institutional aspects is highly relevant. For many items, there are rules that determine their adjustment. Expert projections for administered prices tend to be used for horizons of around four or five quarters. These are longer than for market prices, due to the greater relative relevance of the experts' information.

In the medium-term projection model for administered prices, each item is represented by a specific equation, designed to capture the particularities of its behavior. The definition of these equations considers the specific adjustment rules and measurement methodologies employed by the Brazilian Institute of Geography and Statistics (IBGE). In general, these are not estimated equations, but calibrated equations based on the current institutional framework. The model consists of a set of 24 quarterly equations that describe the idiosyncratic dynamics of these items and are integrated with the small-scale semi-structural models used by the BCB.

In comparison with the model released in 2017, the seasonalities and parameter values were recalibrated to better reflect current conditions. In addition, methodological changes were made to the equations for some items. For example, in the equation for residential electricity prices, the specification based on econometric estimation was replaced by a version calibrated according to the weight of each distributor in the IPCA and

1/ The last release of this model was in the box [Revision of the medium-term projection models for administered prices](#) in the September 2017 IR.

2/ Except for ethanol, which is part of the "industrial goods" segment in the IPCA.

3/ See box [BCB's analysis and projection system](#) in the March 2023 IR.

the weight of Itaipu in each distributor. In the case of fuels, the equations remain similar to those in the previous model but now consider that the ICMS depends only on the IPCA and is only readjusted in the first quarter of each year, and that ethanol and biodiesel prices follow producer prices for gasoline and diesel oil, respectively. Some explanatory variables were also eliminated to make the model more parsimonious.

The administered prices models are continuously revised. Modifications are introduced incrementally as changes in adjustment rules items are identified, ensuring greater adherence to the dynamics of price make up. This continuous approach contributes to greater accuracy in projections and the alignment of analyses with current conditions.

Below is a brief description of the equations for the products with the greatest weight in the IPCA. Next, the impulse response functions are shown in relation to shocks to oil prices, to the exchange rate, to market prices, and to the IVDA. Finally, the detailed equations corresponding to each administered item are presented.

A) Description of the equations of selected items

Residential electricity (weight 3.78% in the IPCA)

Electricity providers have their ordinary tariff adjustments⁴ set by the Brazilian Electric Power National Agency (ANEEL) every year in the anniversary month of the concession contract. ANEEL calculates the adjustment based on the evolution of non-manageable costs, called “Share A”, and manageable costs, called “Share B”, both related to the distributors’ operation. The share of manageable costs is updated by some inflation index (usually IPCA or IGP-M). The share of non-manageable costs is made up of the cost of energy, sector charges (the largest of which is defined annually in the CDE⁵), and transmission costs.⁶

The change in energy costs depends on the type of contract: regular, long-term contracts are usually adjusted annually by inflation indexes; Itaipu energy has its price in USD adjusted annually for the subsequent year (Itaipu energy is around 8% of the energy used by captive consumers, imposing an exchange rate pass-through of around 8% on the energy cost share); and the short-term market, used to adjust differences between contracted energy and effective consumption, is remunerated by the Settlement Price (PLD).⁷

The final tariff adjustment reflects the sum of the increase of the costs of Parcels A and B, with any adjustments to compensate for differences between the projections and the values that effectively occurred in the previous period (known as “financial effects”).

The more complex components of electricity price make up, such as transmission costs and sector charges, are evaluated by the expert over the shorter-term projection horizon. The medium-term modeling for electricity therefore adopts a more simplified approach. In this model, the share of energy from Itaipu allocated to each state is considered, based on their specific contracts, while the rest of the price change is updated using the IPCA accumulated over the previous four quarters. In addition, the applied aggregate seasonality reflects the weight of each electricity provider and the weight of residential electricity in each metropolitan region that makes up the IPCA.

4/ In addition to the ordinary annual adjustment, there are (i) extraordinary adjustments, applied in exceptional situations that lead to an economic and financial imbalance in the contracts, and (ii) a tariff review, which takes place every four or five years and re-evaluates the distributors’ operating costs and investments. These two other adjustment hypotheses are not considered in the model.

5/ Economic Development Account (CDE, in Portuguese). It is a sector charge mainly designed to promote public policies in the electricity sector, such as universal access to energy, subsidies for low-income consumers, and financing for renewable sources.

6/ Part of the transmission costs is calculated annually by ANEEL, and the other part is adjusted in line with past inflation, usually IPCA or IGP-M.

7/ Settlement Price (PLD, in Portuguese), which has maximum and minimum limits set annually by ANEEL and which depends on the hourly scarcity of energy.

In addition to adjustments in electricity tariffs, the IPCA also captures tariff flags.⁸ The model described here refers only to the ordinary tariff share. The flag surcharge is evaluated in the short term by the expert. For the medium term, the pattern of the last four quarters of the expert's short-term projection is repeated seasonally, so that the accumulated projections of electricity tariffs over four quarters are not affected.

The equations used and their parameters are presented at the end of this box.

Pharmaceutical products (weigh 3.46% in the IPCA)

Drug price adjustments are regulated by the Medicines Market Regulation Chamber (CMED), which is linked to the National Health Surveillance Agency (Anvisa). CMED establishes a Maximum Consumer Price (PMC) table, and this table is adjusted annually in April using a known methodology.

The adjustment is calculated as:

$$VPP = IPCA - X + Y + Z$$

where VPP is the percentage change in price ("adjustment" of pharmaceutical products); IPCA is the rate accumulated over 12 months up to the announcement of the adjustment; Factor X is the productivity factor passed on to the consumer⁹; Factor Z is the intra-sectors relative price adjustment factor.¹⁰ Factor Y is the factor that adjusts prices between sectors.¹¹

For the sake of parsimony, the influence of factors X and Z is left out of the medium-term projections. These are variables that are difficult to predict and represent in a medium-term model, remembering that for shorter terms the projection used is that of the expert, who can make projections for these factors using a broader set of information. The Factor Y is calculated according to the following formula, which is a linear simplification of the factor's adjustment rule:

$$\begin{aligned} \text{Factor } Y_t &= c_{\text{cambio}} \left(\sum_{i=1}^4 \Delta C\grave{a}mbio_{t-i} + \sum_{i=1}^4 \Delta CPI_{t-i} - \sum_{i=1}^4 \Delta IPCA_{t-i} \right) \\ &+ c_{\text{energia}} \left(\sum_{i=1}^4 \Delta energia_ex_bandeira_{t-i} - \sum_{i=1}^4 \Delta IPCA_{t-i} \right) \end{aligned}$$

Thus, the projected adjustment is the 12-month IPCA change forecast for the immediately preceding quarter plus Factor Y. Elements of the seasonality of the pass-through over the quarters are also considered. The quarter with the greatest impact is the second of the year.

The change in the price of pharmaceutical products is then modeled as:

$$\begin{aligned} \pi_t^{\text{Produtos farmac\^euticos}} &= d_{1,t} * 0,1 * reajuste_{t-3} + d_{2,t} * 0,8 * reajuste_t + d_{3,t} * 0,05 * reajuste_{t-1} + d_{4,t} * 0,05 * reajuste_{t-2} \end{aligned}$$

where $d_{1,t}$ to $d_{4,t}$ are quarterly dummies and the constants represent the seasonality of the pass-through of the increase in the price of pharmaceutical products. The adjustment is considered to be fully passed on until the first quarter of the subsequent year.

8/ Tariff flags are an additional charge on the electricity tariff to signal to consumers the increase in the cost of energy in that month. They are set in the week before the month of application and vary according to the intensity of the shortage. In general, they can add between 2% and 10% to the basic electricity tariff.

9/ It is always greater than or equal to zero, so Factor X always does not change or reduces the calculated adjustment. Factor X is calculated by the Ministry of Finance's Secretariat for Economic Reforms based on projections of productivity gains by companies producing medicines (for details of the calculation, see the annex to the CMED Resolution 1/2015).

10/ Sectors with a higher share of generic drugs are considered more competitive and will have a higher Z Factor.

11/ For the full calculation, see the annex to the CMED Resolution 1/2015.

Health insurance (weight of 4.06% in the IPCA)

According to the IBGE Technical Note 1/2005¹², to calculate inflation for the “health insurance” sub-item, the IBGE conducts a monthly survey of the most representative health insurance companies in each region, to find out the percentage of adjustment applied to the monthly fees of the contracts that have their anniversary in the month of the survey. In practice, however, the maximum adjustment set by the National Supplementary Health Agency (ANS)¹³ for individual and family health insurance is informed.¹⁴ This annual adjustment is usually published between May and June, for application over the subsequent 12 months, according to the following formula:

$$Reajuste_A = 0,8 * IVDA_A + 0,2 * \Delta IPCA_{A-1}^{ex-plano de saúde}$$

where the subscript A represents the year in question, $IVDA_A$ is the IVDA calculated for year A, and $\Delta IPCA_{A-1}^{ex-plano de saúde}$ is the change in the previous year of the IPCA excluding health insurance. The IVDA calculation is based on the combination of the components Variation in Assistance Expenses (VDA), Efficiency Gains Factor (FGE), and Variation in Revenue by Age Group (VFE).

The IBGE incorporates this adjustment linearly in the IPCA over the subsequent 12 months, representing the fact that adjustments are applied in the anniversary month of each contract, distributed uniformly throughout the year by hypothesis.

In correspondence, in the quarterly model for health insurance inflation, one quarter of the adjustment is applied in the second quarter of the year, and the other three quarters throughout the subsequent three quarters, as in the equation below.

$$\pi_t^{plano saúde} = d_{1,t} \left(\frac{Reajuste_{A-1}}{4} \right) + \sum_{i=2}^4 d_{i,t} \frac{Reajuste_A}{4} + \varepsilon_t^{plano saúde}$$

For projecting the IVDA of the subsequent year, the expert’s judgment is used. Hence, the IVDA is represented in a simplified way in the model with a combination of inertia and indexation to the IPCA, according to the following equation of yearly periodicity:

$$IVDA_A = 0,6 * IVDA_{A-1} + (1 - 0,6) \Delta IPCA_{A-1} + \varepsilon_A^{IVDA}$$

Fuels (total weight 6.96% of the IPCA)

As in the previous versions, consumer fuel prices are projected considering the evolution of the shares that make them up. In general, these shares are the fuel dealer price, added renewable fuel, federal taxes, state taxes, and trade margin.

The behavior of fuel dealer prices is highly dependent on the Petrobras prices policy. Recently, Petrobras has changed its policy of import parity price (IPP) to consider local factors, such as the production cost in Brazil and the Petrobras participation in the market, instead of directly passing through international fluctuations.¹⁵ Nevertheless, international price changes continue to be relevant for the Brazilian price (Figure 1), which justifies maintaining them in the medium-term model.

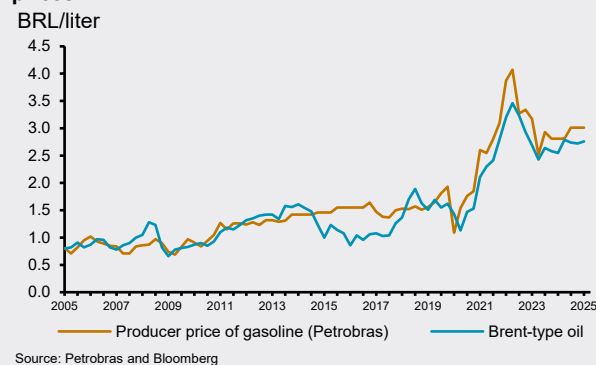
12/ See also [National Consumer Price Index System - Calculation Methods](#), 2020.

13/ The methodology for calculating the maximum adjustment and its components can be found in the annex to CMED Resolution 1/2015 and ANS Normative Resolution 441/2018.

14/ There are also other health insurance plans with a very small impact on the IPCA, called “old” because they were signed before Law 9,656 of 1998, and their adjustments are published in June/July.

15/ See [Petrobras approves gasoline and diesel commercial strategy](#) (Portuguese only), published in May 2023.

Figure 1 – Domestic producer price of gasoline and oil prices



As previously mentioned, short-term projections for fuel prices (and other administered prices) are carried out by the expert, which assesses the convergence of international fuel prices over the oil price, the convergence of Petrobras price over the international fuel price¹⁶ and other relevant factors. For the medium term, it is assumed that the relative prices of Petrobras fuels and the price of gasoline on the international market compared with the price of oil will be in equilibrium and that the changes in Petrobras price will then follow the changes in the international oil price in BRL.

As for added biofuels, it is assumed that prices follow Petrobras prices. For state taxes (ICMS), it is assumed that they are adjusted in the first quarter of each year by the previous year IPCA change. It is also assumed that federal taxes are fixed, and the trade margin follows inflation. Regarding bottled LPG (Liquefied Petroleum Gas), it is assumed that the trade margin grows exclusively in the third quarter of each year, due to the category's adjustment date.

For "vehicle natural gas", a linear regression was estimated between this variable, accumulated inflation over the last four quarters, and changes in oil prices in BRL and the "gasoline" item in the IPCA. A similar strategy was used for "piped natural gas", where the regression considered accumulated inflation over the last four quarters, changes in oil prices in BRL and the "bottled LPG" item in the IPCA.

Vehicle registration and licensing and professional council (2.69% weight at IPCA)

Vehicle registration and licensing and professional council are administered items whose adjustment is known in the first release of the year and is linearly distributed for the subsequent 11 months. For the medium term, it is assumed only past inflation for the adjustment projection. Therefore, equations for both items are the inflation of the previous year divided by four, applied in the first quarter and repeated in the subsequent three quarters

B) Generic model for the other items

The other items of administered prices were divided into two groups according to the presence or absence of any seasonality in the adjustment rule. Items with seasonality are modeled according to the equation below, where seasonal constants come from a combination of estimation with a sample starting in 2020 and judgment.

^{16/} In the sector's terminology, the convergence of the crack spread and of the lag over the PPI.

$$\pi_t^{itens\ com\ sazonalidade} = \sum_{j=1}^4 d_{j,t} c_j * \left(\sum_{i=1}^4 \Delta IPCA_{t-i} \right)$$

For the other items, the equation below is used:

$$\pi_t^{itens\ sem\ sazonalidade} = \left(\frac{1}{4} \right) * \sum_{i=1}^4 \Delta IPCA_{t-i}$$

C) Impulse response functions

This section illustrates the response of administered prices inflation to shocks, using impulse response functions. In particular, shocks to four variables were chosen: exchange rate, oil price, market prices inflation, and IVDA. The first three variables are important components in the modeling of various administered prices and therefore have a more widespread impact, while the last is a very specific item with more limited effects than the others. To fully capture the effect of a shock, even on a specific variable, it is necessary to consider feedback effects. Therefore, to compute the impulse response functions mentioned above, the administered prices model is combined with small-scale semi-structural models, both with the aggregate and the disaggregate models.¹⁷ The weights of administered items in the IPCA have been calibrated according to the values observed in the IPCA of May 2025, released by the IBGE on June 10, 2025.

In fact, as there is a seasonal pattern in the adjustments of a large share of administered prices, the response depends on the quarter when it occurs. For the sake of simplicity, the choice was to present impulse response functions of the linear version of the administered prices model, which summarize the model's behavior and allow understanding the shocks' transmission channels.

The impulse response functions presented below were computed without monetary policy response, i.e., with the Selic rate kept unchanged throughout the entire horizon of the exercise. This procedure allows to clearly visualize the shocks' transmission channels in the absence of a monetary authority's response.

a) Shock on the exchange rate

This exercise verifies what occurs to the projection for administered prices after a permanent increase of 10% in the USD/BRL exchange rate. After four quarters, administered prices reach the maximum effect of around 1.8 p.p. in the four-quarter rate (Figures 2a and 2b). The exchange rate is also passed through to market prices, which reach the maximum effect of around 0.70 p.p. in the four-quarter period, both in the aggregate and disaggregate models. It should be noticed that the impact on administered prices comes especially through the items more sensitive to the exchange rate, such as fuels, medicines, and electricity. As for other items (for instance, health insurance), the effects are indirect through indexation of past inflation. For the IPCA, the 10% exchange rate increase leads to a 1.0 p.p. rise after four quarters.

17/ See box [Updating of small-scale semi-structural models](#) in the June 2024 IR.

Figure 2a – Inflation response to the exchange rate shock

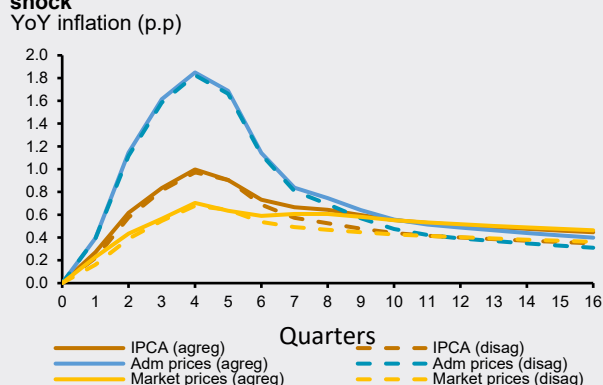
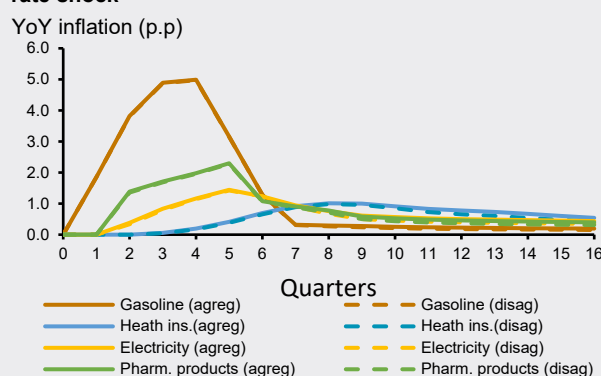


Figure 2b – Selected itens responses to the exchange rate shock



b) Shock on the oil price

A permanent increase of 10% in the Brent-type oil in USD leads to an increase of around 1.3 p.p. in administered prices (Figures 3a and 3b). Oil price affects directly the composition of prices of gasoline, bottled LPG, diesel oil, piped natural gas, and vehicle natural gas. However, the pass-through of each of these prices depends on an individual dynamic. The impact of oil price is stronger on the diesel oil price, since the tax share of diesel oil is lower. However, it is noteworthy that the weight of diesel oil in the IPCA is much lower than that of gasoline (0.25% versus 5.24%). In the fourth quarter, the approximate pass-through is 50% for gasoline and 20% for bottled LPG. The pass-through for bottled LPG is smaller because, given its characteristics, the cost of distribution and trade is proportionally much higher.

Figure 3a – Inflation response to na oil shock

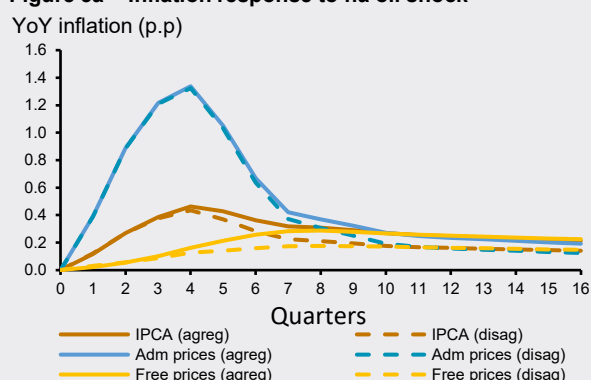
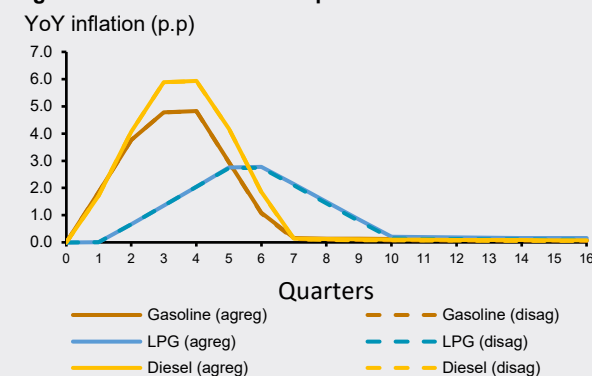


Figure 3b – Selected itens responses to an oil shock



c) Market prices inflation shock

A shock of 1.0 p.p. in market prices inflation also affects administered prices (Figures 4a and 4b). In this case, the shock is applied to the Phillips curve (in the curve of market prices, for the aggregate model, and equally in the three curves of services, industrial goods, and food-at-home, for the disaggregate model). The main transmission channel of market prices inflation to administered prices is indirect through indexation to past inflation. It is noteworthy that the gasoline response is little affected by the shock, as this item is less indexed to past inflation. Conversely, health insurance shows a stronger and more lagged response, due the high inertia of its indexation rule. In general, the indexation mechanisms of administered prices mean that their aggregate index responds more slowly and more persistently than market prices. Five quarters after the shock, when the effect on four-quarter inflation of market prices has already dissipated, the response of inflation in administered prices exceeds that of market prices.

Figure 4a – Inflation response to a shock in market prices

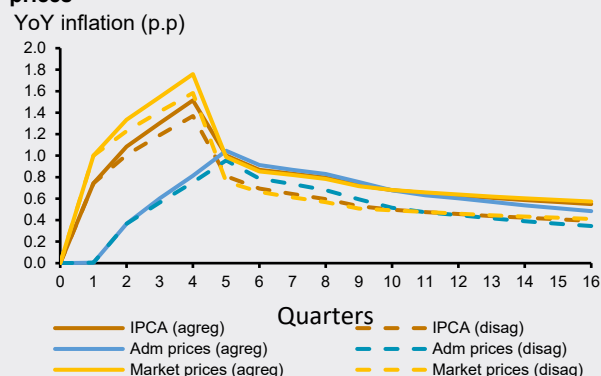
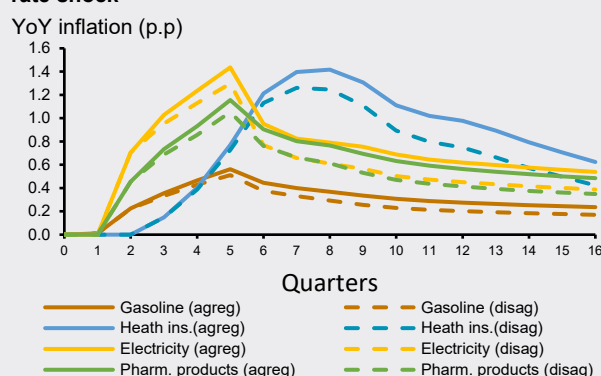


Figure 4b – Selected itens responses to the exchange rate shock



d) IVDA shock

To illustrate how an idiosyncratic shock on a particular item of administered prices spreads to the other IPCA items, an exercise where the IVDA increases 1.0 p.p. is presented. As the IVDA is included in the adjustment rule of health insurance with a weight of 0.8, the four-quarter impact on health insurance prices reaches 0.8 p.p., increasing directly the inflation index of administered prices (increase of 0.14 p.p.) and indirectly the inflation index of market prices (maximum of 0.02 p.p. after eight quarters). Therefore, the effect of the shock on the total IPCA is mostly direct, with a maximum impact of 0.04 p.p. after four quarters.

Figure 5a – Inflation response to a IVDA shock

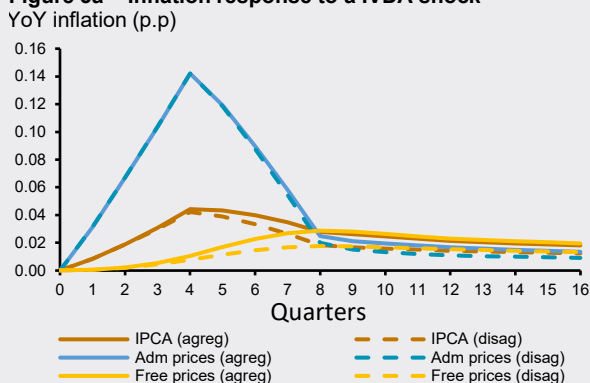
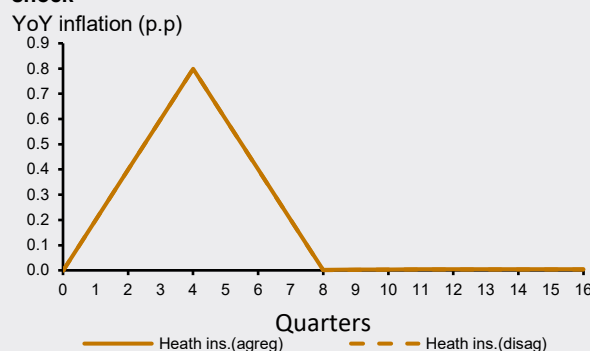


Figure 5b – Health insurance response to na IVDA shock



D) Conclusion

Administered price models presented in this box seek to summarize the dynamics and methodologies of the administered prices composition in Brazil. It is noteworthy that specifications have relevant simplifications for the short term, which are, however, acceptable for the medium-term convergence dynamics. When there is available information for more distant projections, the horizon considered by experts can be extended¹⁸, incorporating these elements in the scenario analysis.

It should be also highlighted that the models described are constantly evolving. Incremental changes can be implemented timely as changes are identified in the adjustment mechanisms and calculation methodologies. This continuous process makes it possible to incorporate new data and adjust the specifications to reflect the best current conditions to build consistent and relevant projection scenarios.

18/ Or possibly shortened if the evaluation indicates that there is no informational advantage for the expert.

Summary table: Equations for administered prices

The table below also gives, along with the name of the administered item, its respective code and weight in the composition of the IPCA. The weight is the value released for the IPCA in May 2025.

Administered item	Equation	Parameters used
Gasoline (cod. 5104001) (weight 5.24%)	$\Delta Petrobras_t = \sum_{i=0}^3 c_i (\Delta Petroleo_{t-i} + \Delta Câmbio_{t-i}) + \varepsilon_t^{\Delta Petrobras}$ $\Delta Etanol_t = \Delta Petrobras_t + \varepsilon_t^{\Delta Etanol}$ $\Delta ICMS_t = d_{1,t} * \sum_{i=1}^4 \Delta IPCA_{t-i} + \varepsilon_t^{\Delta ICMS}$ $\Delta Impostos\ federais_t = \varepsilon_t^{\Delta Impostos\ federais}$ $\Delta Margem_t = \frac{1}{4} * \left[\left(\frac{2}{3} \right) \sum_{i=1}^4 \Delta IPCA_{t-i} + \left(\frac{1}{3} \right) \sum_{i=0}^3 \Delta IPCA_{t-i} \right] + \varepsilon_t^{\Delta Margem}$ $\pi_t^{Gasolina} = p_1 * \Delta Petrobras_t + p_2 * \Delta Etanol_t + p_3 * \Delta ICMS_t + p_4 * \Delta Impostos\ federais_t + p_5 * \Delta Margem_t + \varepsilon_t^{\pi^{Gasolina}}$	$c_0 = 0,4$ $c_1 = 0,4$ $c_2 = 0,2$ $c_3 = 1 - c_0 - c_1 - c_2$ $p_1 = 0,35$ $p_2 = 0,13$ $p_4 = 0,11$ $p_5 = 0,17$ $p_3 = 1 - p_1 - p_2 - p_4 - p_5$
Health insurance (cod. 6203) (weight 4.06%)	$IVDA_A = a * IVDA_{A-1} + (1 - a) IPCA_{A-1} + \varepsilon_A^{IVDA}$ $Reajuste_A = 0,8 * IVDA_A + 0,2 * \Delta IPCA_{A-1}^{ex\ plano\ saude}$ $\pi_t^{Plano\ saude} = d_{1,t} \left(\frac{Reajuste_{A-1}}{4} \right) + \sum_{i=2}^4 d_{i,t} \frac{Reajuste_A}{4}$	$a = 0,6$
Residential electricity (cod. 2202003) (weight 3.78%)	$\pi_t^{Energia} = \pi_t^{Energia\ ex-band} + \pi_t^{bandeira}$ $\pi_t^{bandeira} = -(\pi_{t-1}^{bandeira} + \pi_{t-2}^{bandeira} + \pi_{t-3}^{bandeira}) + \varepsilon_t^{bandeira}$ $\pi_t^{Energia\ ex-band} = (1 - \alpha_{Itaipu} \alpha_{energia}) * \left(\sum_{j=1}^4 d_{j,t} c_j \sum_{i=1}^4 \Delta IPCA_{t-i} \right) + \alpha_{Itaipu} \alpha_{energia} * \sum_{j=1}^4 d_{j,t} c_{Itaipu_j} \sum_{i=1}^4 (\Delta Itaipu_{t-i} + \Delta Câmbio_{t-i}) + \varepsilon_t^{Energia\ ex-band}$ $\Delta Itaipu_t = d_{1,t} * \sum_{i=1}^4 \Delta CPI_{t-i}^{USA} + \varepsilon_t^{\Delta Itaipu}$	$\alpha_{Itaipu} = 0,086$ $\alpha_{energia} = 0,6$ $c_1 = 0,08$ $c_2 = 0,40$ $c_3 = 0,42$ $c_4 = 1 - c_1 - c_2 - c_3$ $c_{Itaipu_1} = 0,1$ $c_{Itaipu_2} = 0,35$ $c_{Itaipu_3} = 0,42$ $c_{Itaipu_4} = 1 - c_{Itaipu_2} - c_{Itaipu_3} - c_{Itaipu_4}$

Pharmaceutical products (cod. 6101) (weight 3.46%)	$Fator X_t = \varepsilon_t^{FatorX}$ $Fator Z_t = \varepsilon_t^{FatorZ}$ $Fator Y_t = \left[c_{cambio} \left(\sum_{i=1}^4 \Delta C\grave{a}mbio_{t-i} + \sum_{i=1}^4 \Delta CPI_{t-i}^{USA} - \sum_{i=1}^4 \Delta IPCA_{t-i} \right) + c_{energia} \left(\sum_{i=1}^4 \pi_{t-i}^{Energia\ ex-b\grave{a}nd} - \sum_{i=1}^4 \Delta IPCA_{t-i} \right) \right] + \varepsilon_t^{FatorY}$ $reajuste_t = \sum_{i=1}^4 \Delta IPCA_{t-i} - Fator X_t + Fator Y_t + Fator Z_t$ $\pi_t^{produtos\ farmac\^euticos} = d_{1,t} c_{rep,1} * reajuste_{t-3} + d_{2,t} c_{rep,2} * reajuste_t + d_{3,t} c_{rep,3} * reajuste_{t-1} + d_{4,t} c_{rep,4} * reajuste_{t-2} + \varepsilon_t^{produtos\ farmac\^euticos}$	$c_{C\grave{a}mbio} = 0,15$ $c_{Energia} = 0,07$ $c_{rep,1} = 0,1$ $c_{rep,2} = 0,8$ $c_{rep,3} = 0,05$ $c_{rep,4} = 1 - c_{rep,1} - c_{rep,2} - c_{rep,3}$
Vehicle registration and license fee (cod. 5102004) (weight 2.69%)	$\pi_t^{Emplacamento} = d_{1,t} * \left(\frac{1}{4} \right) * \sum_{i=1}^4 \Delta IPCA_{t-i} + \sum_{i=2}^4 d_{i,t} \pi_{t-1}^{Emplacamento} + \varepsilon_t^{Emplacamento}$	
Water and sewage fee (cod. 2101004) (weight 1.84%)	$\pi_t^{Taxa\ \acute{a}gua\ e\ esgoto} = \sum_{j=1}^4 d_{j,t} c_j \sum_{i=1}^4 \Delta IPCA_{t-i} + \varepsilon_t^{Taxa\ \acute{a}gua\ e\ esgoto}$	$c_1 = 0,19$ $c_2 = 0,50$ $c_3 = 0,09$ $c_4 = 1 - c_1 - c_2 - c_3$
Bottled LPG (cod. 2201004) (weight 1.25%)	$\Delta Petrobras_t^{GLP} = \sum_{i=0}^5 c_i (\Delta Petroleo_{t-i} + \Delta C\grave{a}mbio_{t-1}) + \varepsilon_t^{Petrobras, GLP}$ $\Delta ICMS_t^{botij\tilde{a}o} = d_{1,t} * \sum_{i=1}^4 \Delta IPCA_{t-i} + \varepsilon_t^{ICMS, botij\tilde{a}o}$ $\Delta Impostos\ federais_t^{botijao} = \varepsilon_t^{ImpFed, botij\tilde{a}o}$ $\Delta Margem_t^{botij\tilde{a}o} = d_{3,t} * \left[\left(\frac{2}{3} \right) \sum_{i=1}^4 \Delta IPCA_{t-i} + \left(\frac{1}{3} \right) \sum_{i=0}^3 \Delta IPCA_{t-i} \right] + \varepsilon_t^{\Delta Margem, botij\tilde{a}o}$ $\pi_t^{G\acute{a}s\ botij\tilde{a}o} = p_1 * \Delta Petrobras_t^{GLP} + p_3 * \Delta ICMS_t^{botij\tilde{a}o} + p_4 * \Delta Impostos\ federais_t^{botijao} + p_5 * \Delta Margem_t^{botij\tilde{a}o} + \varepsilon_t^{\pi^{botijao}}$	$c_0 = 0,0$ $c_1 = 0,2$ $c_2 = 0,2$ $c_3 = 0,2$ $c_4 = 0,2$ $c_5 = 0,2$ $p_1 = 0,32$ $p_4 = 0$ $p_5 = 0,51$ $p_3 = 1 - p_1 - p_4 - p_5$
Urban bus (cod. 5101001) (weight 1.12%)	$\pi_t^{\hat{o}nibus\ urbano} = \sum_{j=1}^4 d_{j,t} c_j \sum_{i=1}^4 \Delta IPCA_{t-i} + \varepsilon_t^{\pi^{\hat{o}nibus\ urbano}}$	
Lottery (cod. 7201063) (weight 0.44%)	$\pi_t^{logos\ de\ azar} = \left(\frac{1}{4} \right) * \sum_{i=1}^4 \Delta IPCA_{t-i} + \varepsilon_t^{\pi^{logos\ de\ azar}}$	

Inter-municipal bus (cod. 5101006) (weight 0.41%)	$\pi_t^{\hat{\text{ônibus intermunicipal}}} = \sum_{j=1}^4 d_{j,t} c_j \sum_{i=1}^4 \Delta IPCA_{t-i} + \varepsilon_t^{\pi^{\hat{\text{ônibus intermunicipal}}}}$	$c_1 = 0,55$ $c_2 = 0,14$ $c_3 = 0,18$ $c_4 = 1 - c_1 - c_2 - c_3$
Diesel oil (cod. 5104003) (weight 0.25%)	$\Delta Petrobras_t^{diesel} = \sum_{i=0}^3 c_i (\Delta Petroleo_{t-i} + \Delta Câmbio_{t-1}) + \varepsilon_t^{\Delta Petrobras^{diesel}}$ $\Delta Biodiesel_t = \Delta Petrobras_t$ $\Delta ICMS_t^{diesel} = d_{1,t} * \sum_{i=1}^4 \Delta IPCA_{t-i} + \varepsilon_t^{\Delta ICMS^{diesel}}$ $\Delta Impostos federais_t^{diesel} = \varepsilon_t^{\Delta Impostos federais^{diesel}}$ $\Delta Margem_t^{diesel} = \left(\frac{1}{4}\right) * \sum_{i=1}^4 \Delta IPCA_{t-i} + \varepsilon_t^{\Delta Margem^{diesel}}$ $\pi_t^{\hat{\text{ôleo diesel}}} = p_1 * \Delta Petrobras_t + p_2 * \Delta Biodiesel_t + p_3 * \Delta ICMS_t + p_4 * \Delta Impostos federais_t + p_5 * \Delta Margem_t + \varepsilon_t^{\pi^{\hat{\text{ôleo diesel}}}}$	$c_0 = 0,3$ $c_1 = 0,4$ $c_2 = 0,3$ $c_3 = 1 - c_0 - c_1 - c_2$ $p_1 = 0,468$ $p_2 = 0,125$ $p_4 = 0,05$ $p_5 = 0,184$ $p_3 = 1 - p_1 - p_2 - p_4 - p_5$
Land-line telephone plan (cod. 9101002) (weight 0.22%)	$\pi_t^{\text{Plano telefonia fixa}} = \left(\frac{1}{4}\right) * \sum_{i=1}^4 \Delta IPCA_{t-i} + \varepsilon_t^{\pi^{\text{Plano telefonia fixa}}}$	
Taxi (cod. 5101002) (weight 0.20%)	$\pi_t^{\text{Taxi}} = \sum_{j=1}^4 d_{j,t} c_j \sum_{i=1}^4 \Delta IPCA_{t-i} + \varepsilon_t^{\pi^{\text{Taxi}}}$	$c_1 = 0,48$ $c_2 = 0,31$ $c_3 = 0,04$ $c_4 = 1 - c_1 - c_2 - c_3$
Piped natural gas (cod. 2201005) (weight 0.15%)	$\pi_t^{\text{Gás encanado}} = \beta_1 \frac{1}{4} \left(\sum_{i=1}^4 \Delta IPCA_{t-i} \right) + \beta_2 \pi_{t-1}^{\text{Gás de botijão}} + \beta_3 \sum_{i=1}^4 (\Delta Petrôleo_{t-i} + \Delta Câmbio_{t-i}) + \varepsilon_t^{\pi^{\text{Gás encanado}}}$	$\beta_1 = 0,55$ $\beta_2 = 0,25$ $\beta_3 = 0,05$
Interstate bus (cod. 5101007) (weight 0.11%)	$\pi_t^{\hat{\text{ônibus interestadual}}} = \sum_{j=1}^4 d_{j,t} c_j \sum_{i=1}^4 \Delta IPCA_{t-i} + \varepsilon_t^{\pi^{\hat{\text{ônibus interestadual}}}}$	$c_1 = 0,17$ $c_2 = 0,02$ $c_3 = 0,06$ $c_4 = 1 - c_1 - c_2 - c_3$
Traffic fines (cod. 5102006) (weight 0.088%)	$\pi_t^{\text{Multa}} = \left(\frac{1}{4}\right) * \sum_{i=1}^4 \Delta IPCA_{t-i} + \varepsilon_t^{\pi^{\text{Multa}}}$	

Toll (cod. 5102015) (weight 0.087%)	$\pi_t^{Pedágio} = \sum_{j=1}^4 d_{j,t} c_j \sum_{i=1}^4 \Delta IPCA_{t-i} + \varepsilon_t^{Pedágio}$	$c_1 = 0,22$ $c_2 = 0,04$ $c_3 = 0,42$ $c_4 = 1 - c_1 - c_2 - c_3$
Vehicle natural gas (cod. 5104005) (weight 0.070%)	$\begin{aligned} \pi_t^{Gás\ veicular} &= \beta_1 \left(\sum_{i=1}^4 \Delta IPCA_{t-i} \right) + \beta_2 \pi_{t-1}^{Gasolina} + \beta_3 \pi_{t-2}^{gasolina} \\ &+ \sum_{i=1}^3 \gamma_i (\Delta Petr\acute{o}leo_{t-i} + \Delta C\grave{a}mbio_{t-i}) + \varepsilon_t^{Gás\ veicular} \end{aligned}$	$\beta_1 = 0,5$ $\beta_2 = 0,1$ $\beta_3 = 0,2$ $\gamma_1 = 0,1$ $\gamma_2 = 0,05$ $\gamma_3 = 0,05$
Subway (cod. 5101011) (weight 0.066%)	$\pi_t^{Metr\hat{o}} = \sum_{j=1}^4 d_{j,t} c_j \sum_{i=1}^4 \Delta IPCA_{t-i} + \varepsilon_t^{Metr\hat{o}}$	$c_1 = 0,38$ $c_2 = 0,58$ $c_3 = 0,04$ $c_4 = 1 - c_1 - c_2 - c_3$
Postal services (cod. 9101011) (weight 0.065%)	$\pi_t^{Correio} = \sum_{j=1}^4 d_{j,t} c_j \sum_{i=1}^4 \Delta IPCA_{t-i} + \varepsilon_t^{Correio}$	$c_1 = 0,17$ $c_2 = 0,83$ $c_3 = 0$ $c_4 = 1 - c_1 - c_2 - c_3$
Integrated public transport (cod. 5101053) (weight 0.052%)	$\pi_t^{Integra\c{c}\~{a}o} = d_{1,t} \sum_{i=1}^4 \Delta IPCA_{t-i} + \varepsilon_t^{integracao}$	
Professional council (cod. 7101090) (weight 0.048%)	$\begin{aligned} \pi_t^{Conselho\ de\ classe} &= \frac{1}{4} d_{1,t} \sum_{i=1}^4 \Delta IPCA_{t-i} + \sum_{i=2}^4 d_{i,t} \pi_{t-1}^{Conselho\ de\ classe} + \varepsilon_t^{Conselho\ de\ classe} \end{aligned}$	
Train (cod. 5101004) (weight 0.038%)	$\pi_t^{Trem} = d_{1,t} \sum_{i=1}^4 \Delta IPCA_{t-i} + \varepsilon_t^{trem}$	
Notary (cod. 7101034) (weight 0.023%)	$\pi_t^{Cart\acute{o}rio} = d_{1,t} \sum_{i=1}^4 \Delta IPCA_{t-i} + \varepsilon_t^{cartorio}$	